Power BI Assignment 2 – DAX & Data Visualization on the E-Commerce Sales Dataset



🔍 **Project Objective:**

★ The objective of this project is to analyse e-commerce sales data using Power BI and DAX by calculating key business metrics such as order count, average profit, year-to-date (YTD) sales, profit margin, and sales target achievement.

★ It aims to create interactive visualizations to compare actual sales with targets, track monthly sales trends, evaluate product profitability, and analyse sales performance across categories, sub-categories, cities, and states.

★ The final goal is to identify sales and profit gaps and recommend practical actions to improve business performance.

📂 **Dataset Description:**

The dataset covers the period from 1 April 2018 to 31 March 2019.

- Total orders: 500
- Order line items: 1,500
- Cities: 24
- States: 19
- Categories: 3
- Sub-categories: 17

❖ The project uses three datasets.

Dataset	Rows	Purpose
List of Orders	500	Contains order date, customer, city, and state details.
Order Details	1,500	Contains sales amount, profit, quantity, and product details.
Sales Target	36	Contains monthly sales targets for each category.

🗂️ **Column Description:**

🛒 **List of Orders:**

Column Name	Description
Order ID	Unique ID of the order (primary key)
Order Date	Date the order was placed
Customer Name	Name of the customer
State	State of delivery
City	City of delivery

📦 **Order Details:**

Column Name	Description
Order ID	Order the line item belongs to (foreign key)
Amount	Sales amount of the line item
Profit	Profit (or loss, if negative) of the line item
Quantity	Units sold
Category	Product category (Clothing, Electronics, Furniture)
Sub-Category	Product sub-category (e.g. Printers, Saree, Tables)

🎯 **Sales Target:**

Column Name	Description
Month of Order Date	Month of the target (Apr-18 … Mar-19)
Category	Product category
Target	Sales target for that category and month

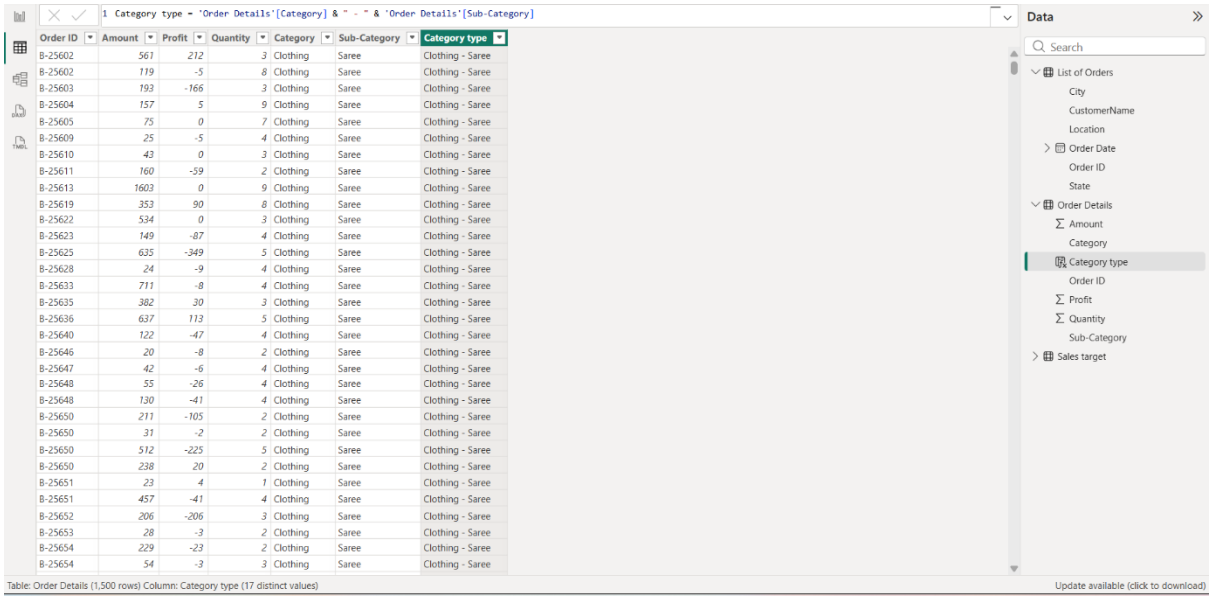
DAX Calculations

Calculated Columns:

I. Create a Calculated Column for 'Category Type:**

Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.

```
Category type = 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]
```

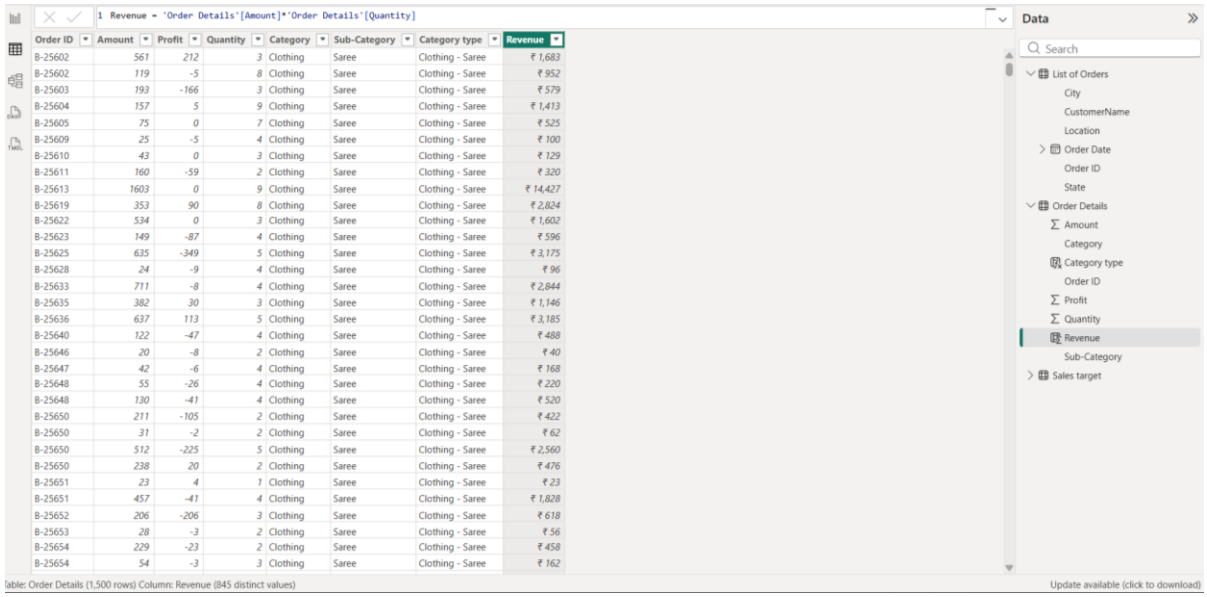


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type
B-25602	561	212	3	Clothing	Saree	Clothing - Saree
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree
B-25604	157	5	9	Clothing	Saree	Clothing - Saree
B-25605	75	0	7	Clothing	Saree	Clothing - Saree
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree
B-25610	43	0	3	Clothing	Saree	Clothing - Saree
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree
B-25619	353	90	8	Clothing	Saree	Clothing - Saree
B-25622	534	0	3	Clothing	Saree	Clothing - Saree
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree
B-25635	382	30	3	Clothing	Saree	Clothing - Saree
B-25636	637	113	5	Clothing	Saree	Clothing - Saree
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree
B-25650	238	20	2	Clothing	Saree	Clothing - Saree
B-25651	23	4	1	Clothing	Saree	Clothing - Saree
B-25651	457	-41	4	Clothing	Saree	Clothing - Saree
B-25652	206	-206	3	Clothing	Saree	Clothing - Saree
B-25653	28	-3	2	Clothing	Saree	Clothing - Saree
B-25654	229	-23	2	Clothing	Saree	Clothing - Saree
B-25654	54	-3	3	Clothing	Saree	Clothing - Saree

II. Calculate Revenue per Order in Order Details Table:

Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.

```
Revenue = 'Order Details'[Amount]*'Order Details'[Quantity]
```

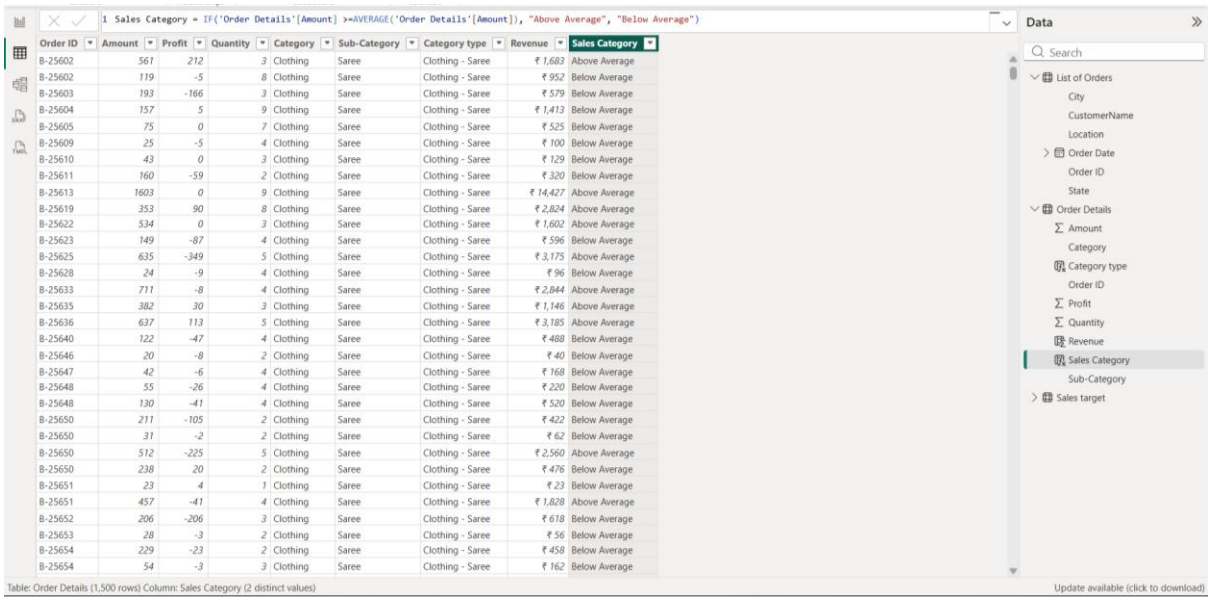


Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type	Revenue
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	₹ 1,683
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	₹ 952
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	₹ 579
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	₹ 1,413
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	₹ 525
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	₹ 100
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	₹ 129
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	₹ 320
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	₹ 14,427
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	₹ 2,824
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	₹ 1,602
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	₹ 596
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	₹ 3,175
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	₹ 96
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	₹ 2,844
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	₹ 1,146
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	₹ 3,185
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	₹ 488
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	₹ 40
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	₹ 168
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	₹ 220
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	₹ 520
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	₹ 422
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree	₹ 62
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree	₹ 2,560
B-25650	238	20	2	Clothing	Saree	Clothing - Saree	₹ 476
B-25651	23	4	1	Clothing	Saree	Clothing - Saree	₹ 23
B-25651	457	-41	4	Clothing	Saree	Clothing - Saree	₹ 1,828
B-25652	206	-206	3	Clothing	Saree	Clothing - Saree	₹ 618
B-25653	28	-3	2	Clothing	Saree	Clothing - Saree	₹ 56
B-25654	229	-23	2	Clothing	Saree	Clothing - Saree	₹ 458
B-25654	54	-3	3	Clothing	Saree	Clothing - Saree	₹ 162

III. Create a Calculated Column to Categorize Sales:

Add a calculated column named ‘Sales Category’ in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

```
Sales Category = IF ('Order Details'[Amount] >= AVERAGE ('Order Details'[Amount]), "Above Average", "Below Average")
```



Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category type	Revenue	Sales Category
B-25602	561	212	3	Clothing	Saree	Clothing - Saree	₹ 1,683	Above Average
B-25602	119	-5	8	Clothing	Saree	Clothing - Saree	₹ 952	Below Average
B-25603	193	-166	3	Clothing	Saree	Clothing - Saree	₹ 579	Below Average
B-25604	157	5	9	Clothing	Saree	Clothing - Saree	₹ 1,413	Below Average
B-25605	75	0	7	Clothing	Saree	Clothing - Saree	₹ 525	Below Average
B-25609	25	-5	4	Clothing	Saree	Clothing - Saree	₹ 100	Below Average
B-25610	43	0	3	Clothing	Saree	Clothing - Saree	₹ 129	Below Average
B-25611	160	-59	2	Clothing	Saree	Clothing - Saree	₹ 320	Below Average
B-25613	1603	0	9	Clothing	Saree	Clothing - Saree	₹ 14,427	Above Average
B-25619	353	90	8	Clothing	Saree	Clothing - Saree	₹ 2,824	Above Average
B-25622	534	0	3	Clothing	Saree	Clothing - Saree	₹ 1,602	Above Average
B-25623	149	-87	4	Clothing	Saree	Clothing - Saree	₹ 596	Below Average
B-25625	635	-349	5	Clothing	Saree	Clothing - Saree	₹ 3,175	Above Average
B-25628	24	-9	4	Clothing	Saree	Clothing - Saree	₹ 96	Below Average
B-25633	711	-8	4	Clothing	Saree	Clothing - Saree	₹ 2,844	Above Average
B-25635	382	30	3	Clothing	Saree	Clothing - Saree	₹ 1,146	Above Average
B-25636	637	113	5	Clothing	Saree	Clothing - Saree	₹ 3,185	Above Average
B-25640	122	-47	4	Clothing	Saree	Clothing - Saree	₹ 488	Below Average
B-25646	20	-8	2	Clothing	Saree	Clothing - Saree	₹ 40	Below Average
B-25647	42	-6	4	Clothing	Saree	Clothing - Saree	₹ 168	Below Average
B-25648	55	-26	4	Clothing	Saree	Clothing - Saree	₹ 220	Below Average
B-25648	130	-41	4	Clothing	Saree	Clothing - Saree	₹ 520	Below Average
B-25650	211	-105	2	Clothing	Saree	Clothing - Saree	₹ 422	Below Average
B-25650	31	-2	2	Clothing	Saree	Clothing - Saree	₹ 62	Below Average
B-25650	512	-225	5	Clothing	Saree	Clothing - Saree	₹ 2,560	Above Average
B-25650	238	20	2	Clothing	Saree	Clothing - Saree	₹ 476	Below Average
B-25651	23	4	1	Clothing	Saree	Clothing - Saree	₹ 23	Below Average
B-25651	457	-41	4	Clothing	Saree	Clothing - Saree	₹ 1,828	Above Average
B-25652	206	-206	3	Clothing	Saree	Clothing - Saree	₹ 618	Below Average
B-25653	28	-3	2	Clothing	Saree	Clothing - Saree	₹ 56	Below Average
B-25654	229	-23	2	Clothing	Saree	Clothing - Saree	₹ 458	Below Average
B-25654	54	-3	3	Clothing	Saree	Clothing - Saree	₹ 162	Below Average

Calculated Measures

I. Order Count:

Create a measure that calculates the total number of orders in the Order Details table.

```
Order Count = DISTINCTCOUNT ('Order Details'[Order ID])
```



II. Average Profit in Delhi:

Create a measure that calculates the average profit for orders placed in Delhi.

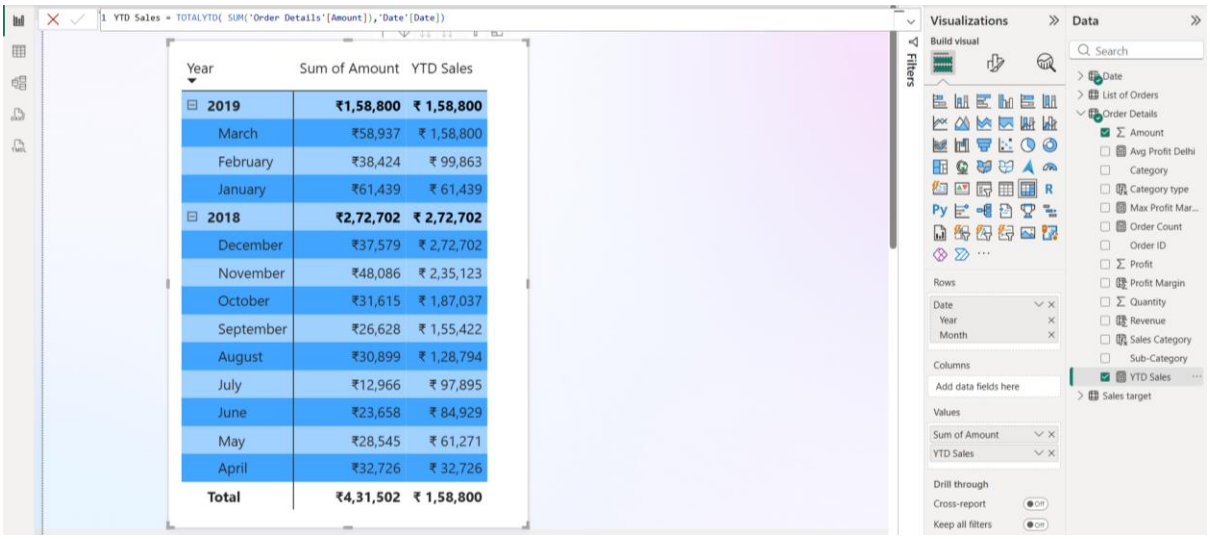
Average Profit Delhi = CALCULATE (AVERAGE ('Order Details'[Profit]), 'List of Orders'[State] = "Delhi")



III. Year-to-Date (YTD) Sales:

Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.

YTD Sales = TOTALYTD (SUM ('Order Details'[Amount]),'Date'[Date])



Data Visualization

Create the following visualizations using Power BI

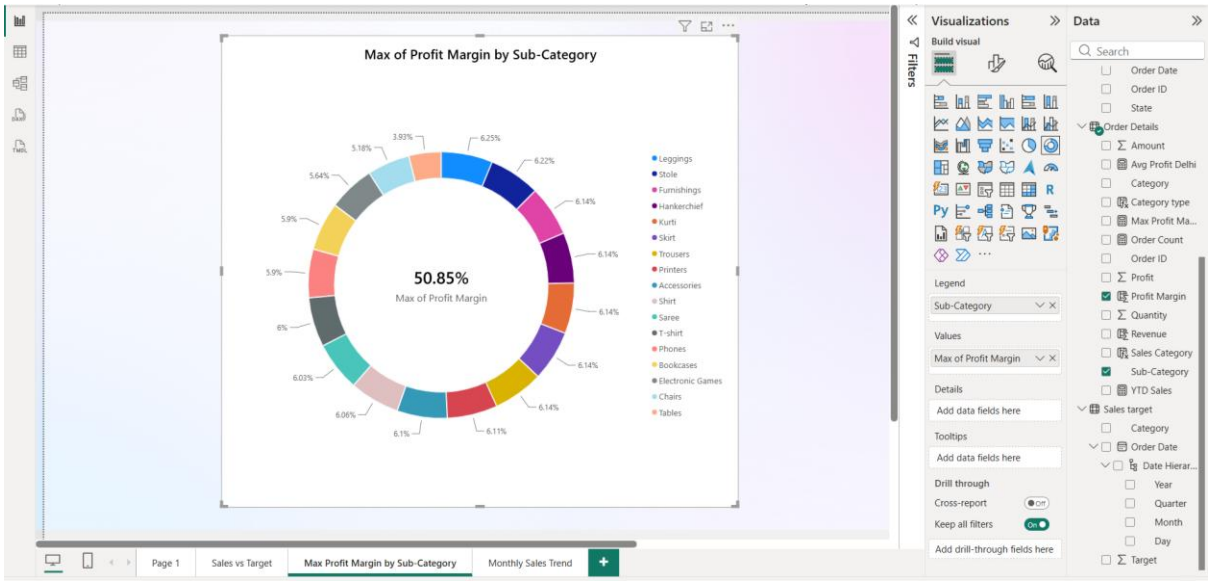
I. Sales Target Achievement by Category:

Compare actual sales with sales targets by category using a clustered column chart.



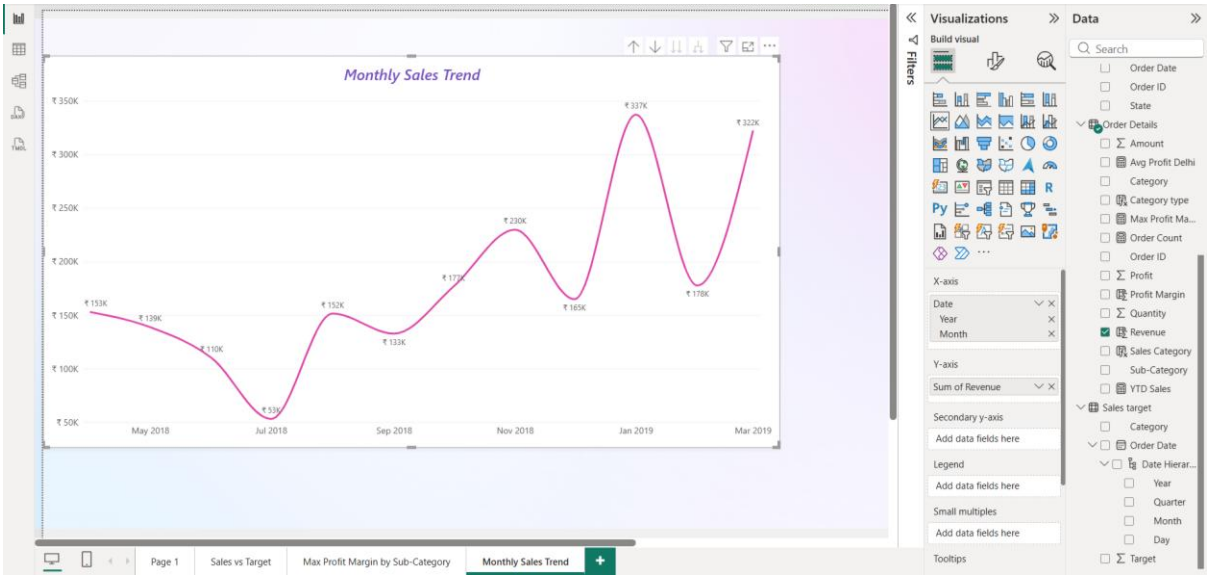
II. Max Profit Margin by Sub-Category:

Analyse the maximum profit margin for each sub-category of products using a donut chart.



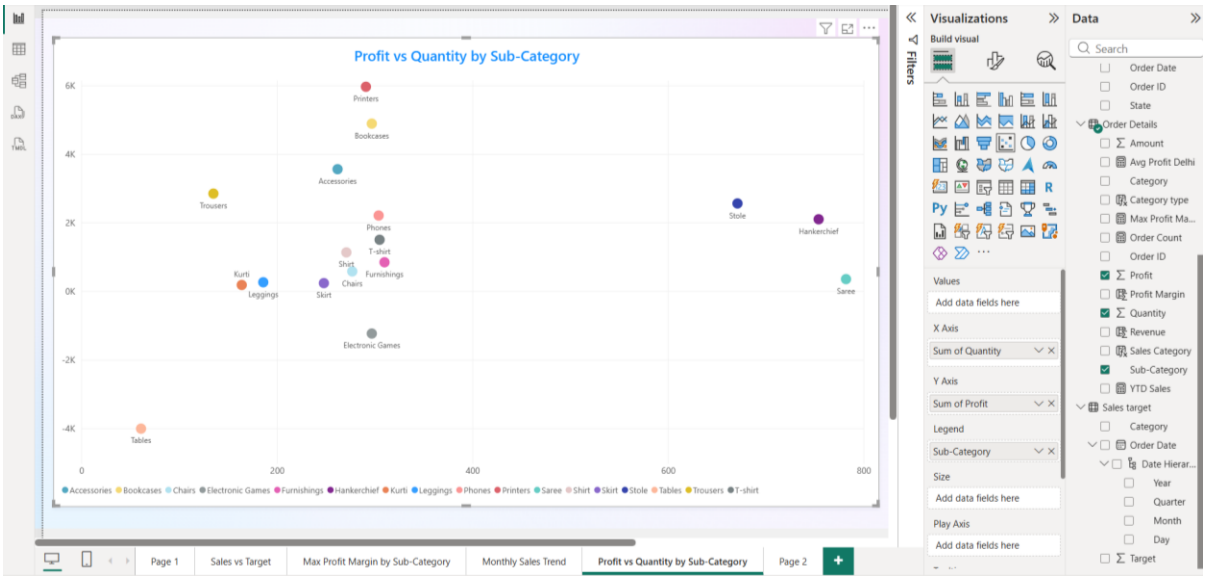
III. Monthly Sales Trend:

Show the trend of monthly sales over time using a line chart.



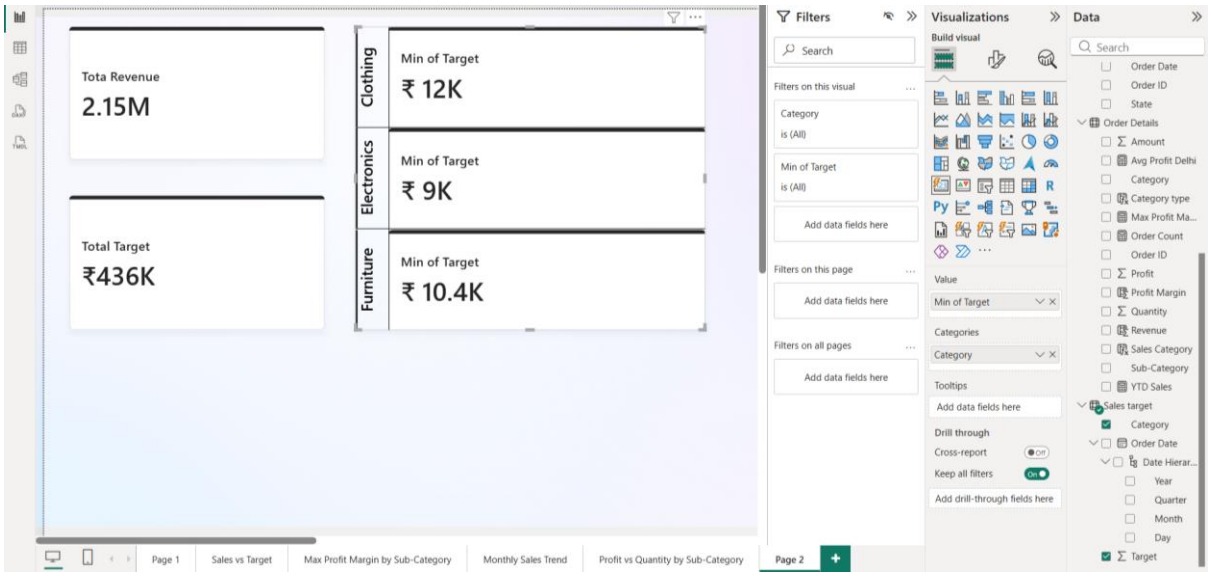
IV. Comparison of Profit and Quantity by Sub-Category:

Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.



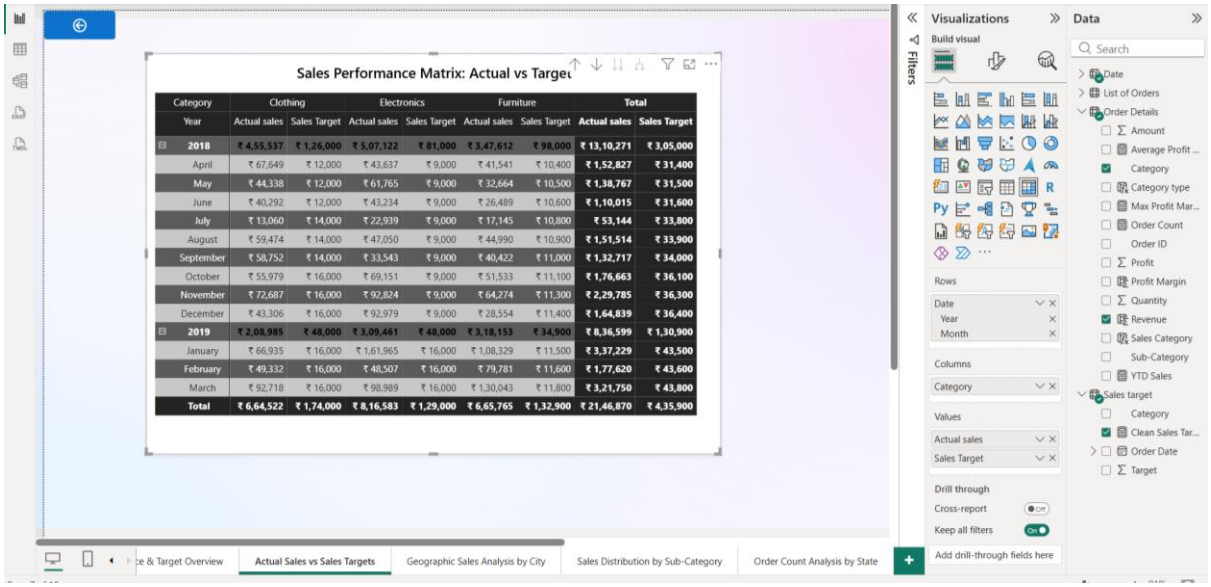
V. Comparison of Total Sales Amount and Target:

Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.



VI. Sales Performance Matrix:

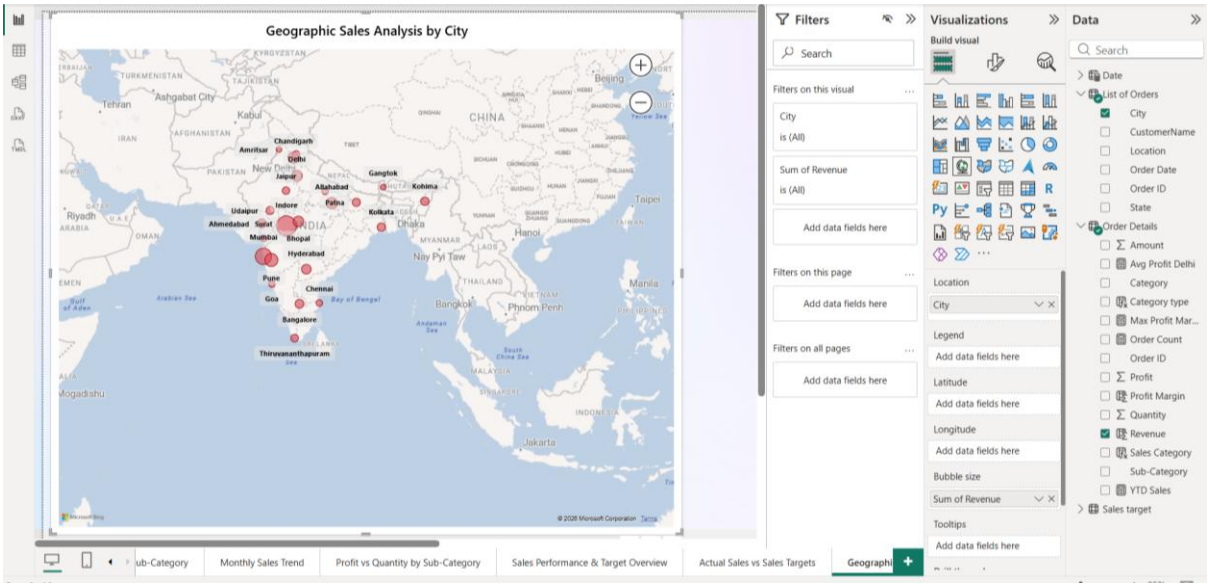
Build a matrix view to analyse how actual sales compare to sales targets across different categories and months.



Category	Clothing		Electronics		Furniture		Total	
Year	Actual sales	Sales Target	Actual sales	Sales Target	Actual sales	Sales Target	Actual sales	Sales Target
2018	₹ 4,55,537	₹ 1,26,000	₹ 5,07,122	₹ 81,000	₹ 3,47,612	₹ 98,000	₹ 13,10,271	₹ 3,05,000
April	₹ 67,649	₹ 12,000	₹ 43,637	₹ 9,000	₹ 41,541	₹ 10,400	₹ 1,52,827	₹ 31,400
May	₹ 44,338	₹ 12,000	₹ 61,765	₹ 9,000	₹ 32,664	₹ 10,500	₹ 1,38,767	₹ 31,500
June	₹ 40,292	₹ 12,000	₹ 43,234	₹ 9,000	₹ 26,489	₹ 10,600	₹ 1,10,015	₹ 31,600
July	₹ 13,060	₹ 14,000	₹ 22,939	₹ 9,000	₹ 17,145	₹ 10,800	₹ 53,144	₹ 33,800
August	₹ 59,474	₹ 14,000	₹ 47,050	₹ 9,000	₹ 44,990	₹ 10,900	₹ 1,51,514	₹ 33,900
September	₹ 58,752	₹ 14,000	₹ 33,543	₹ 9,000	₹ 40,422	₹ 11,000	₹ 1,32,717	₹ 34,000
October	₹ 55,979	₹ 16,000	₹ 69,151	₹ 9,000	₹ 51,533	₹ 11,100	₹ 1,76,663	₹ 36,100
November	₹ 72,687	₹ 16,000	₹ 92,824	₹ 9,000	₹ 64,274	₹ 11,300	₹ 2,29,785	₹ 36,300
December	₹ 43,306	₹ 16,000	₹ 92,979	₹ 9,000	₹ 28,554	₹ 11,400	₹ 1,64,839	₹ 36,400
2019	₹ 2,08,985	₹ 48,000	₹ 3,09,461	₹ 48,000	₹ 3,18,153	₹ 34,900	₹ 8,36,599	₹ 1,30,900
January	₹ 66,935	₹ 16,000	₹ 1,61,965	₹ 16,000	₹ 1,08,329	₹ 11,500	₹ 3,37,229	₹ 43,500
February	₹ 49,332	₹ 16,000	₹ 48,507	₹ 16,000	₹ 79,781	₹ 11,600	₹ 1,77,620	₹ 43,600
March	₹ 92,718	₹ 16,000	₹ 98,989	₹ 16,000	₹ 1,30,043	₹ 11,800	₹ 3,21,750	₹ 43,800
Total	₹ 6,64,522	₹ 1,74,000	₹ 8,16,583	₹ 1,29,000	₹ 6,65,765	₹ 1,32,900	₹ 21,46,870	₹ 4,35,900

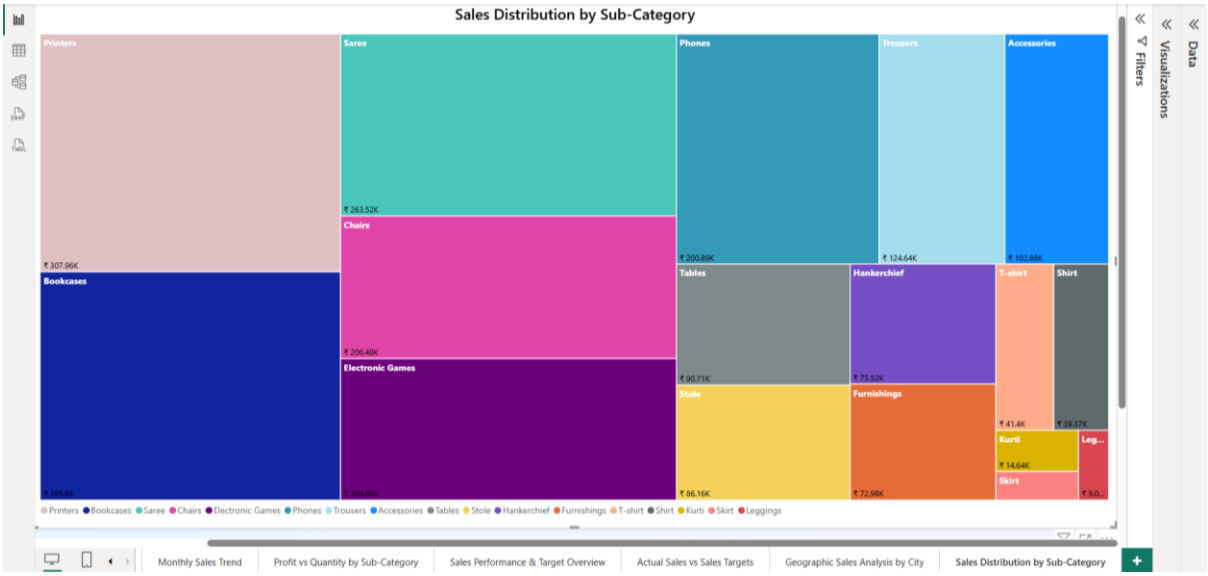
VII. Geographic Sales Analysis:

Visualize total sales on a map by city to identify regional sales patterns.



VIII. Sales Distribution by Sub-Category:

Represent the sales distribution across different sub-categories using a tree map.

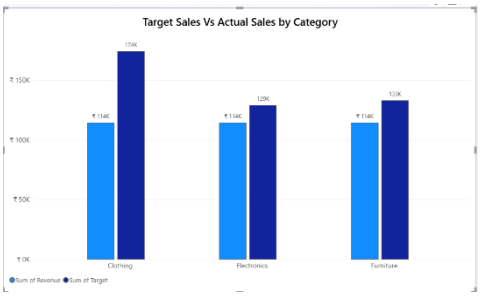


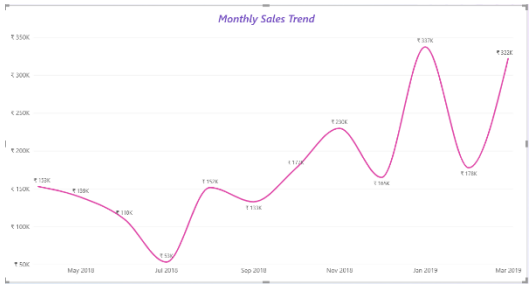
IX. Order Count Analysis by State:

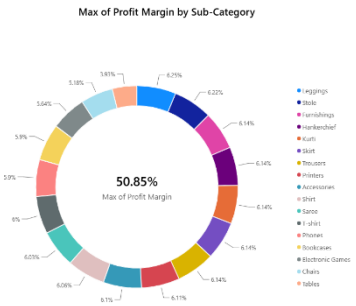
Create a funnel chart to visualize the distribution of order counts across different states.

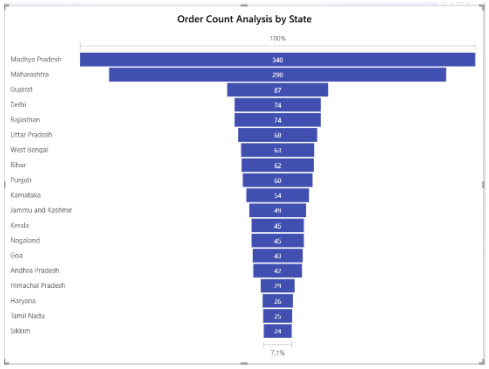


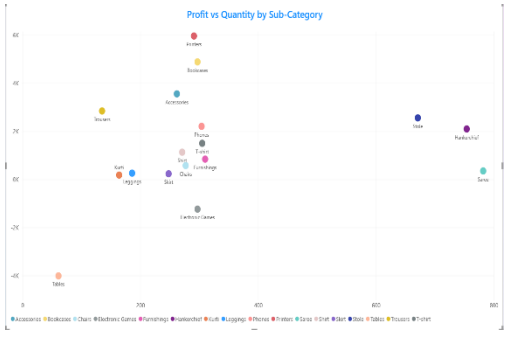
Visualization, Dashboard & Business Insights:



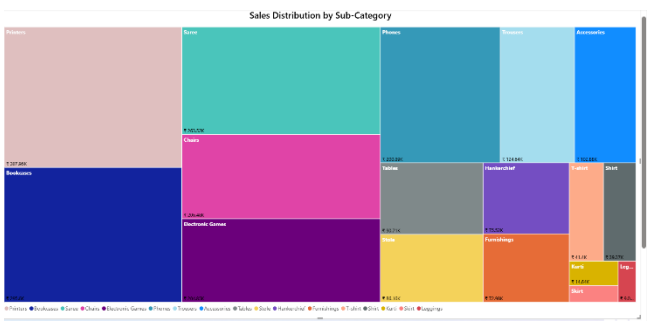












Sales Performance Matrix: Actual vs Target

Category	Clothing		Electronics		Furniture		Total	
Year	Actual sales	Sales Target	Actual sales	Sales Target	Actual sales	Sales Target	Actual sales	Sales Target
2018	₹ 4,55,537	₹ 1,26,000	₹ 5,07,122	₹ 81,000	₹ 3,47,612	₹ 98,000	₹ 13,10,271	₹ 3,05,000
April	₹ 67,649	₹ 12,000	₹ 43,637	₹ 9,000	₹ 41,541	₹ 10,400	₹ 1,52,827	₹ 31,400
May	₹ 44,338	₹ 12,000	₹ 61,765	₹ 9,000	₹ 32,664	₹ 10,500	₹ 1,38,767	₹ 31,500
June	₹ 40,292	₹ 12,000	₹ 43,234	₹ 9,000	₹ 26,489	₹ 10,600	₹ 1,10,015	₹ 31,600
July	₹ 13,060	₹ 14,000	₹ 22,939	₹ 9,000	₹ 17,145	₹ 10,800	₹ 53,144	₹ 33,800
August	₹ 59,474	₹ 14,000	₹ 47,050	₹ 9,000	₹ 44,990	₹ 10,900	₹ 1,51,514	₹ 33,900
September	₹ 38,752	₹ 14,000	₹ 33,543	₹ 9,000	₹ 40,422	₹ 11,000	₹ 1,32,717	₹ 34,000
October	₹ 55,979	₹ 16,000	₹ 69,151	₹ 9,000	₹ 51,533	₹ 11,100	₹ 1,76,663	₹ 36,100
November	₹ 72,687	₹ 16,000	₹ 92,824	₹ 9,000	₹ 64,274	₹ 11,300	₹ 2,29,785	₹ 36,300
December	₹ 48,306	₹ 16,000	₹ 92,979	₹ 9,000	₹ 28,554	₹ 11,400	₹ 1,64,839	₹ 36,400
2019	₹ 2,08,985	₹ 48,000	₹ 3,09,461	₹ 48,000	₹ 3,18,153	₹ 34,900	₹ 8,36,599	₹ 1,30,900
January	₹ 66,935	₹ 16,000	₹ 1,61,965	₹ 16,000	₹ 1,08,329	₹ 11,500	₹ 3,37,229	₹ 43,500
February	₹ 49,332	₹ 16,000	₹ 48,507	₹ 16,000	₹ 79,781	₹ 11,600	₹ 1,77,620	₹ 43,600
March	₹ 92,718	₹ 16,000	₹ 98,989	₹ 16,000	₹ 1,30,043	₹ 11,800	₹ 3,21,750	₹ 43,800
Total	₹ 6,64,522	₹ 1,74,000	₹ 8,16,583	₹ 1,29,000	₹ 6,65,765	₹ 1,32,900	₹ 21,46,870	₹ 4,35,900

🔍 Descriptive Analysis:

- Total revenue was approximately ₹2.15 million.
- Electronics recorded the highest sales, followed by Clothing and Furniture.
- January 2019 had the highest monthly sales, while July 2018 had the lowest.
- Madhya Pradesh and Maharashtra had the highest order counts.

🧩 Diagnostic Analysis:

- Electronics performed well compared to the other categories.
- Clothing had high order quantities but did not meet its sales target.
- Tables and some other products generated losses, reducing overall profit.
- Sales varied across months and product categories.

🔮 Predictive Analysis:

- Electronics may continue to perform well if the trend continues.
- Clothing and Furniture may need improvement to increase sales.
- Sales may be higher during peak months such as January and March.
- Loss-making products may continue to affect profit.

💡 Prescriptive Analysis:

- Promote products that generate higher revenue and profit.
- Review the pricing of loss-making products such as Tables.
- Improve sales performance in Clothing and Furniture.
- Plan promotions during low-sales months to increase revenue.

✅ Conclusion

- This project demonstrates how Power BI, DAX, and data visualization can be used to analyse e-commerce sales data and understand business performance.
- The calculated columns, measures, and nine visualizations helped identify revenue trends, sales target achievement, category performance, and profitability issues.
- The analysis highlighted top-performing categories, loss-making products, and areas that require improvement.
- These insights can help the business make better decisions by improving product pricing, reducing losses, promoting profitable products, setting realistic sales targets, and planning campaigns during peak months.
- Overall, the project shows how data analysis can transform raw sales data into meaningful business insights and support better decision-making to improve future revenue and profitability.